Ferns & Petals (FNP)
Sales Analytics Project

Excel dashboard • Business questions • Data preparation • Portfolio report

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Project type: Sales and customer analytics using a supplied sample dataset

Executive summary

This project examines 1,000 order records alongside product and customer reference tables. The goal is to organize transaction data into useful views of sales value, monthly performance, product demand, customer geography, gifting occasions, and delivery duration. Using quantity multiplied by listed price as a working sales-value definition, the dataset totals ■3,520,984. The result should be treated as an analytical estimate until the metric definition is confirmed against the original business rules.

1. Business context

Gift purchases are often linked to specific events such as birthdays, anniversaries, festivals, and seasonal celebrations. Looking at sales by date, product, city, and occasion can help a retailer plan stock, understand demand peaks, and review the delivery experience. This project translates the supplied order records into a set of practical business questions rather than treating the spreadsheet as a collection of unrelated rows.

2. Objectives

- Estimate overall sales value and average spend per order row.
- Summarize the time between order and delivery.
- Compare sales performance across months in 2023.
- Identify products with the highest estimated sales value.
- Rank cities by order volume.
- Compare sales value across gifting occasions.
- Explore whether larger quantities tend to align with longer delivery durations.
- Present key measures and comparisons in a readable Excel dashboard.

3. Data used

The workbook contains customer, order, and product tables. The order table includes order and delivery dates/times, product and customer IDs, quantity, location, occasion, month/hour helper fields, a supplied delivery-duration field, and listed price. The product table contains product name, category, price, and occasion. The customer table contains location and demographic fields as well as direct personal identifiers; those identifiers are excluded from the sanitized files prepared for public sharing.

4. Approach

1. Inspect the tables and check data types, missing values, duplicate identifiers, and date fields.
2. Use the product/customer IDs to connect order records with reference information where required.
3. Create a working sales-value field as quantity multiplied by listed price; verify this assumption before presenting it as revenue.
4. Aggregate sales by month, product, city, and occasion using PivotTables or equivalent summaries.
5. Calculate average delivery duration from the supplied duration field and inspect how it relates to order quantity.
6. Build KPI cards and charts, then reconcile dashboard totals with the underlying rows.

5. Initial dataset checks

Measure	Value	Interpretation
Order table rows	1,000	Rows in supplied order table; verify distinct Order_ID count.
Estimated sales value	■3,520,984	Sum of Quantity × Price (INR).
Average quantity per row	3.04	Mean quantity recorded per order row.
Average delivery duration	5.53 days	Mean of supplied diff_order_delivery field.
Estimated value per row	■3,520.98	Estimated sales value divided by row count.

These values are recalculated from the supplied workbook using an explicit working definition. They are not audited company results. Before publishing, check whether price is per unit, whether order IDs are unique, and whether the delivery-duration field is measured in calendar days.

6. Dashboard design

KPI cards: Estimated sales value, average delivery duration, average quantity, and order-row count.

Monthly trend: A month-by-month column or line chart to show changes across 2023.

Product performance: Ranked bars for the top products by estimated sales value.

Geographic demand: Top cities ranked by number of order records.

Occasion comparison: Sales-value comparison across occasions.

Delivery analysis: Scatter plot or grouped comparison of quantity against delivery duration.

7. Business value

The dashboard can support decisions about seasonal inventory, product promotion, city-level marketing, and delivery planning. It also creates a consistent way to discuss performance with non-technical stakeholders. Any action should be based on validated metrics and interpreted in context—for example, high sales value does not necessarily mean high profit if product costs and delivery costs are not available.

8. Limitations and responsible sharing

- The workbook is a supplied sample and should not be described as live or official company reporting.
- Revenue is estimated as quantity multiplied by listed price; confirm the intended business definition.
- Average value per row is not necessarily average order value if one order can have multiple rows.
- An association between quantity and delivery duration does not establish causation.

- Do not publish the original customer table containing names, contact numbers, emails, or full addresses. Use the anonymized CSV included with this project.

9. Conclusion

This project provides a structured Excel analytics workflow for exploring sales, product demand, customer geography, gifting occasions, and delivery performance. Its main strength is the conversion of transaction-level records into focused questions that can guide further investigation. The next improvement is to validate metric definitions, document the final dashboard findings, and keep all public project files free of personal customer information.